

Navigate Advantage Access for Friis Epidemiology 101, Third Edition (Open Enrollment)
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CHAPTER 3 Epidemiologic Measurements Used to Describe Disease Occurrence
Mathematical Terms Used in Epidemiology

Mathematical Terms Used in Epidemiology

Some important mathematical terms applied to epidemiologic measures are rate, proportion, and percentage. These measures are types of ratios. (Refer to **Figure 3.1**.) The following section defines these terms and gives calculation examples of ratios, proportions, and percentages for mortality from COVID-19. The topic of rates will be covered later in the chapter. Data for use in calculating the examples of rates, proportions, and percentages are given in **Table 3.2**.

Background: During one week in an elementary school classroom, 6 boys and 4 girls get a cold			
Ratio (R)	$R = \frac{A}{B}$	A and B can be any two numbers, even ratios themselves.	The ratio of colds in boys to girls is 6/4.
Rate (r)	$r = \frac{A}{\Delta t}$	A ratio where the numerator is a count, and the denominator is a measure of time.	The rate (r) of colds in boys is 6 per week.
Proportion (p)	$p = \frac{A}{A + B}$	A ratio where the numerator is a part of the denominator.	The proportion of colds in boys is 6/10 or 0.6
Percentage (P)	$P = \left(\frac{A}{A + B} \right) \times 100$	A proportion multiplied by 100.	60% of colds in the classroom occurred in boys

Figure 3.1 Definitions of mathematical terms that are used in epidemiology

Description

TABLE 3.2 Data for Calculations of Rates, Proportions, and Percentages

Cumulative U.S. COVID-19 mortality, 2020–2022, all ages ¹	Males = 511,000
	Females = 421,860

Author's hypothetical survey of clinic patients ($n = 20$) regarding electronic cigarette (e-cigarette) use	Number of e-cigarette users = 15 Number of persons who did not use = 5
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Ratio

A **ratio** is the result of dividing one quantity by another. Rates and proportions are examples of ratios. In some situations ratios can be presented as a percentage.² Although a ratio consists of a numerator and a denominator, its most general form does not necessarily have any specified relationship between the numerator and denominator.

A ratio is expressed as follows: ratio = X/Y .
Calculation example of a ratio:

Example 1: With respect to COVID-19 mortality, the sex ratio of deaths (male to female deaths) = X/Y , where: $X = 511,000$ and $Y = 421,860$. The sex ratio = $511,000/421,860 = 1.2$ to 1 (approximately).

Example 2: Referring to the data in Table 3.2, you can observe that the ratio of users of e-cigarettes to nonusers is $15/5$, or 3 to 1.

Example 3: In demography, the sex ratio refers to the number of males per 100 females. In the United States, the sex ratio in 2020 was estimated to be 98.0, meaning that there were more women and girls than men and boys.³ At the same time, there were considerable variations by state; Alaska had the highest male to female ratio, while the District of Columbia had the lowest.³ At birth, the sex ratio is approximately 105 males to 100 females. Due to higher mortality among males, the sex ratio decreases with age, a trend shown in **Figure 3.2**. At the highest age groups, females outnumber males and this pattern has remained consistent over the last 30 years.

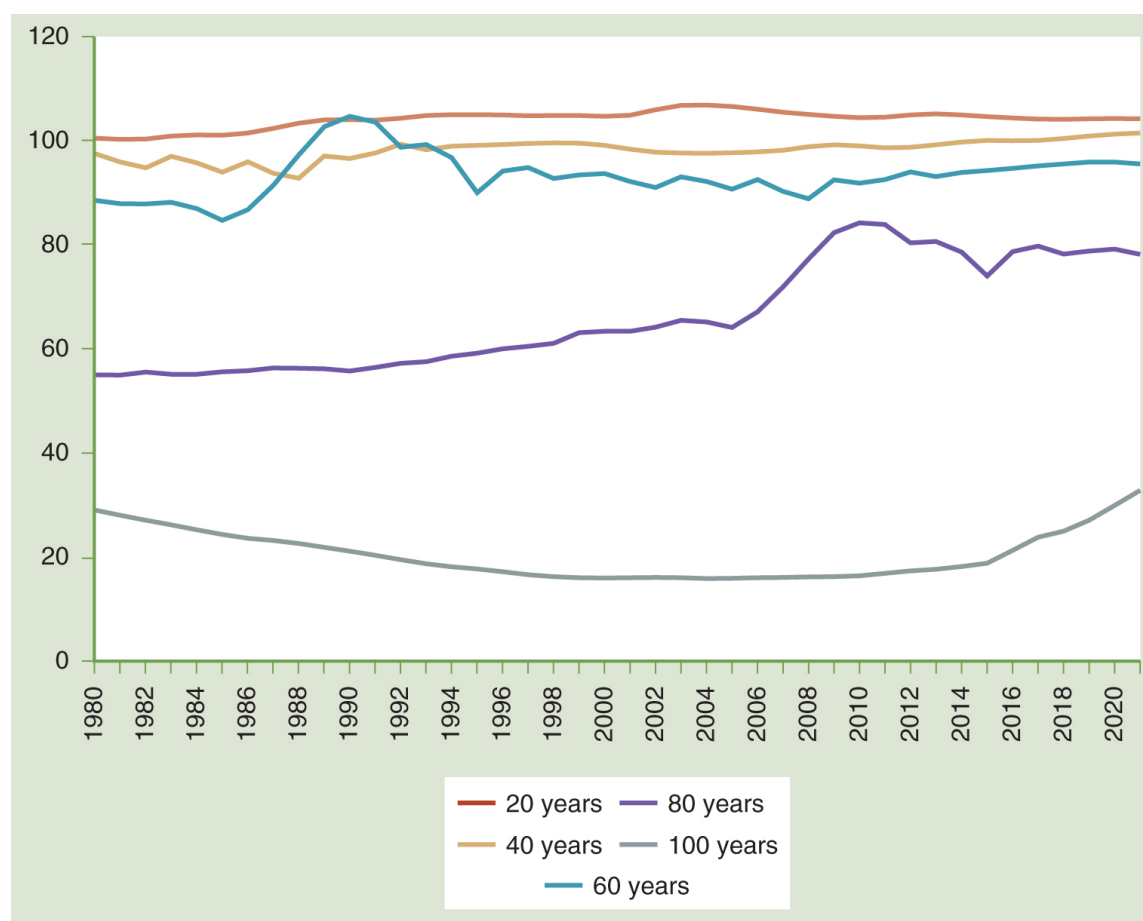


Figure 3.2 Sex ratio, by year and age

Data from United Nations, Department of Economic and Social Affairs, Population Division.
World Population Prospects 2022, Online Edition.

Description

Proportion

A **proportion** is a type of ratio in which the numerator is a subset (or part) of the denominator. Proportions may be expressed as percentages.

A proportion is expressed as follows: $\text{proportion} = A/(A+B)$

Calculation example of a proportion:

Example 1: Proportion of COVID-19 deaths in males

Suppose that A = the number of male deaths from COVID-19

A = 511,000

B = the number of female deaths from COVID-19

B = 421,860

The proportion of deaths that occurred among males = $511,000 / (511,000 + 421,860) = 0.55$

Example 2: Proportion of e-cigarette users (data from Table 3.2)

$$\text{Proportion} = 15 / (15 + 5) = 0.75$$

Percentage

A **percentage** is a proportion that has been multiplied by 100. The formula for a percentage is as follows:

$$\text{percentage} = (A/(A+B)) \times 100$$

Example 1: The percentage of male deaths from COVID-19 was $(0.55 \times 100) = 55\%$.

Example 2: The percentage of e-cigarette users was $(0.75 \times 100) = 75\%$.

Example 3: Refer to **Figure 3.3**, which is a graph of the percentage of office-based physicians who used telemedicine by specialty in 2019 and 2021. The figure demonstrates the large increase in the use of telemedicine from 2019 to 2021, in large part due to the COVID-19 pandemic. We also see some variation in telemedicine use by specialty, with surgeons least likely to use telemedicine.⁴

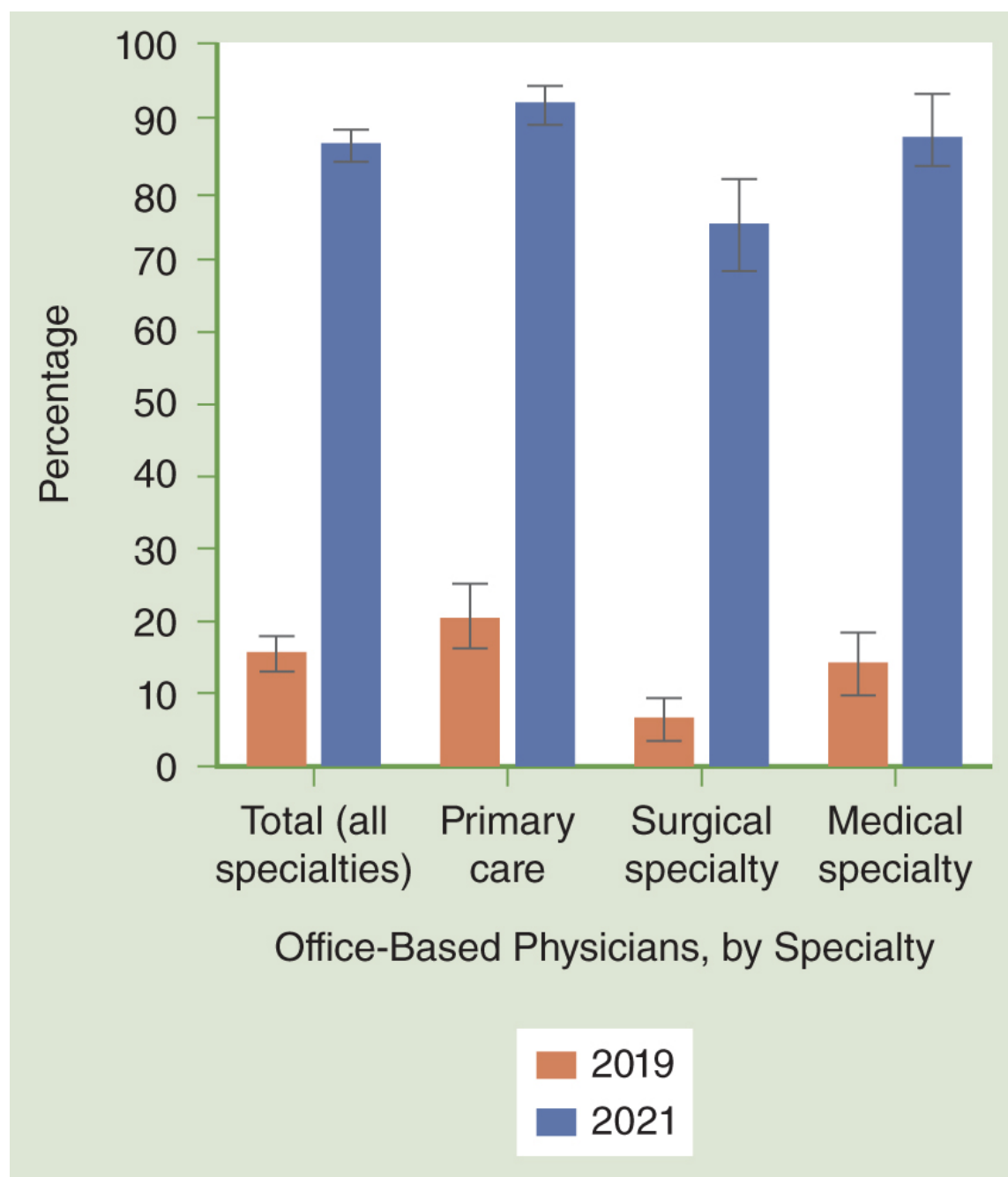


Figure 3.3 Percentage of office-based physicians using telemedicine technology, by specialty—United States, 2019 and 2021

Centers for Disease Control and Prevention. QuickStats: Percentage of office-based physicians using telemedicine technology, by specialty — United States, 2019 and 2021. *MMWR Morb Mortal Wkly Rep.* 2022;71:1565.

Description

Let's consider how a proportion (as well as a percentage) can be helpful in describing health conditions. A proportion indicates how important a health outcome is relative to the size of a group. Refer to the following example: Suppose there were 10 college dorm residents who had infectious mononucleosis (a viral disease that produces fever, sore throat, and

tiredness). How large a problem did these 10 cases represent? To answer this question, one would need to know whether the dormitory housed 20 students or 500 students. If there were only 20 students, then 50% (or 0.50) were ill. Conversely, if there were 500 students in the dormitory, then only 2% (or 0.02) were ill. Clearly, these two scenarios paint a completely different picture of the magnitude of the problem. In this situation, expressing the count as a proportion is indeed helpful. In most situations, it will be informative to have some idea about the size of the denominator. Although the construction of a proportion is straightforward, one of the central concerns of epidemiology is to find and enumerate appropriate denominators to describe and compare groups in a meaningful and useful way.

Rate

A **rate** is another type of ratio and differs from a proportion because the denominator involves a measure of time. (Refer back to Figure 3.1). The rate measure shown in the figure is the mathematical formula in which elapsed time is denoted in the denominator by the symbol Δt .

Epidemiologic rates are composed of a numerator (the number of events such as health outcomes), a denominator (a population in which the events occur), and a measure of time.¹ This measure of time is the period during which events in the numerator occur. The denominator consists of the average population in which the events occurred during this same time period.

In epidemiology, rates are used to measure risks associated with exposures and provide information about the speed of development of a disease. Also, rates can be used to make comparisons between populations. Medical publications may use the terms ratio, proportion, and rate without strict adherence to the mathematical definitions for these terms. Hence, you must be mindful of how a measure is defined and calculated in a specific situation.